Parole Officer Effects on Reentry: Support, Deterrence, and Bias

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Abstract:

Little is known about the effects of the parolee-officer relationship. In this work, I use the quasi-random assignment of paroled individuals to parole officers in Pennsylvania to evaluate the effects of individual officers on reentry success. First, I find that individual officers affect documented parole violations and employment but not future arrests. Leveraging the results of a survey of parole officers, I find that officers that prioritize interpersonal support reduce parole violations while officers that prioritize skill-building and structural support over deterrence have no observable effect on post-release success. Next, I find that paroled individuals assigned to officers of other races or genders experience increased recidivism and decreased employment. However, I find no evidence that this effect is driven by specific officers. Instead, it is detected at a relatively constant rate across all parolee-officer pairings, suggesting it is due to systemic or societal bias that impacts parolee-officer relationships as opposed to "a few bad apples."

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1. Introduction

One out of every 50 U.S. adults is serving a convicted sentence under some form of government supervision. This includes a large number of individuals incarcerated in State and Federal prisons and an even larger number supervised in the community under parole or probation supervision – in 2019, 1.4 million individuals were incarcerated in State and Federal prisons, 0.9 million were living in the community under parole supervision, and 3.5 million were living in the community under probation supervision (Carson 2020; Oudekerk & Kaeble 2021; U.S. Census Bureau 2019). Despite the size of U.S. carceral system, post-conviction outcomes for supervised individuals are quite poor. For individuals released from prison, 71% are re-arrested within five years of release, 46% return to prison within five years of release (Durose & Antenangeli 2021), and 55% are unemployed eight months after release (Visher et al. 2008).

Outcomes are also poor for individuals under community supervision. Recidivism rates are higher for individuals released under parole supervision than for individuals released without it – in Pennsylvania in 2016 the three-year recidivism rate for individuals released to parole was 66.2% relative to 56.8% for individuals released without supervision at the completion of their convicted sentence (Pennsylvania Department of Corrections 2022). Individuals assigned probation supervision also realize poor outcomes – in 2019 36.9% and 31.1% of individuals who exited parole and probation supervision, respectively, exited for unsuccessful reasons such as incarceration or absconding (Oudekerk & Kaeble 2021).

To improve outcomes, policy makers need clear evidence about the effects of community supervision to determine how best to facilitate post-conviction success. Despite the size of the U.S. carceral system and poor post-conviction outcomes, almost no evidence exists on the effects of community supervision (Doleac & LaForest 2022). Further, given the wide variation in community supervision policy across states and municipalities in the United States (Phelps & Curry 2017), a better understanding of which policies and procedures are most effective, and why, is needed to improve these programs and post-release outcomes.

One of the main components of community supervision is one's assigned probation or parole officer. Officers work with individuals assigned to probation or parole as both case workers (to facilitate reentry into the community) and law enforcement officers (to enforce supervisory conditions) (Chamberlain et al. 2017). Over many meetings, officers provide interpersonal support to supervised individuals, build rapport with them, connect them with community programs and

resources, and convey a clear message of deterrence regarding future criminal behavior and violations of supervisory conditions. When overseeing community supervision, officers have a wide degree of discretion in how they build relationships with individuals, structure supervision, and how they respond when individuals break conditions – a decision that can have a large and lasting impact on an individual's future.

This paper explores the role of parole officers in facilitating post-conviction community success. It first investigates the extent to which one's parole officer impacts success. That is, do certain officers reduce recidivism and increase employment for the individuals they are assigned, relative to other officers? It next investigates why certain officers impact post-conviction success. That is, are effects driven by a focus on interpersonal support, structural support, or deterrence? Third, it investigates how demographic ingroup and outgroup differences between officers and paroled individuals impact success. That is, does being assigned an officer of a different race of sex impact one's outcome, and are these outgroup effects driven by individual or societal biases?

This work investigates the effects of parole officers using individual-level data on the Pennsylvania parole population between 2006 and 2020. It estimates the causal effects of parole officers by leveraging the quasi-random assignment of paroled individuals to parole officers. Specifically, in Pennsylvania paroled individuals are first assigned to parole officer units based on their proposed residency location and required special needs (such as sexual offenders and domestic violence offenders). Within each parole officer unit, individuals are assigned to parole officers based on current officer caseload sizes. As such, similar paroled individuals assigned to the same officer unit are assigned to different officers based on slight differences in assignment dates.

I find that, first, individual officers have little relative effect on paroled individuals' future arrests or major parole violations that lead to reincarceration. However, individual parole officers do have a notable effect on paroled individuals' minor parole violations – that result in written warnings or additional parole conditions. Specifically, a one standard deviation increase in officer expected effect on minor parole violations increases the chance a paroled individual receives a minor violation in the first year after release by 5%. Finally, individual officers have a notable effect on parole officer-documented employment but not on tax-record documented employment. This results suggests that officers vary in reporting behavior but have little impact on actually employment success.

Next, leveraging the results of a survey of parole officer perspectives, I find that officers that focus on interpersonal supports – noting that they care about parolees, try to learn about them, and prioritize building good relationships and rapport – reduce minor parole violations. However, officers that focus on skill-building and structural supports as opposed to messages of deterrence – noting that they work with parolees to build pro-social behavior, try to facilitate reentry, as think of themselves more as case workers than law enforcement officers – have little observable effect on minor parole violations. None of the three types of focus (interpersonal support, structural support, or deterrence) appears to have an effect on future arrests or employment.

Finally, I find evidence of racial and gender outgroup disparities in parolee-officer relationships that impacts recidivism, parole violations, and employment. Specifically, being assigned to an officer of a different racial background leads to slight increases in future recidivism (6%) and minor parole violations (3%) and a slight decrease in employment (6%). Being assigned to an officer of a different gender has a similar but slightly stronger effect – an increase in recidivism (7%) and a decrease in employment (8%), but no observable effect on minor parole violations. Further analysis finds that these differences in outcomes occur across nearly all parolee-officer relationships, at a fairly uniform rate, rejecting a hypothesis that these disparities are driven by a subset of specific parole officers. Instead, the results support a hypothesis that the disparities are driven by existing structural and societal bias that impact the relationships paroled individuals and parole officers are able to jointly construct.

Overall, the results imply that relative differences in parole officer approaches have little effect on future offending and reincarceration, regardless of whether officers focus on providing interpersonal support, structural support, or deterrence. However, these differences do affect minor parole violations and officer-documented employment, which often result in additional parole restrictions that increase the requirements of supervision on paroled individuals. For minor violations, officers that prioritize interpersonal supports appear to have most success.

The results also find that racial and gender differences in parole-officer pairings impact recidivism in addition to minor parole violations and employment. As this effect appears to be driven by aggregate structural or societal biases which impacts parolee-officer relationships, these results motive a conclusion that these biases need to be corrected at an external, societal level. However, making parole officers aware of these biases can potentially help to reduce them.

Overall, little research exists on the effects of community supervision. Of the small set of existing studies, most conclude that additional community supervision has little corrective impact (Doleac & LaForest 2022). Regarding probation, several randomized controlled trials find no effect of specific intensive probation programs relative to standard programs (Petersilia & Turner 1993; Lane et al. 2005; Henneguelle et al. 2010; Hyatt & Barnes 2014; Barnes et al. 2012). Regarding parole, recent work by Banan (2022) compares the effects of release to parole supervision relative to release without supervision in North Carolina, using a regression discontinuity design. She finds that parole supervision leads to a short-term increase in parole violations that offsets a slight decrease in new crimes, with no long-term effects. Zapryanova (2020) investigates the effect of additional time spent on parole, using the random assignment of judges in Georgia along with discontinuities in parole board guidelines, and finds no effect of additional time under parole supervision on returns to prison.

Next, Georgiou (2014) investigates the effects of parole supervision intensity levels in Washington State, using a regression discontinuity design around two risk score cutoffs that determine supervision intensity level, and finds no effect of higher supervision levels at either margin. Finally, LaForest-Tucker (2026) separately identifies the effects of parole supervision intensity levels and special conditions such as curfew, placement in a halfway house, or enrollment in a drug treatment program using two discontinuities in risk score cutoffs that determine supervision level and the random assignment of parole hearing voters that determine special conditions in Pennsylvania. He finds that, along most margins, increased supervision leads to additional parole violations with little effect on future arrests or employment.

Only one prior study looks at the causal effects of parole officers. Specifically, Andersen and Wildeman (2015) leverage the random assignment of parolees to parole officers in Copenhagen and find that officers have an individual effect on dependency on public benefits and recidivism, but not employment. All other existing studies on the effects of parole officers is correlational or qualitative. Bares & Mowen (2019) conduct a matching analysis using a panel data survey and find a correlation between officers that provide professional support and decreased recidivism, but no correlation between interpersonal support and recidivism. Morash et al. (2016) find no effect of individual officers on female parolees, using a matching analysis. Finally, Chamberlin et al. (2018) leverage a survey of parolees and find that those that mentioned having better relationships with their parole officers had better parole outcomes.

A separate set of research focuses on the effects of racial bias in the criminal justice system. Existing research largely focuses on how racial bias impacts police officer and judge decision-making, in one-off interactions with potential offenders. For example, Arnold et al. (2018) find evidence of racial bias in judge bail decisions in Miami and Philadelphia. Weisburst (2024) finds variation on arrest propensity across officers in Dallas, and Gonclaves & Mello (2021) find that minorities are less likely to receive discounts on their speeding tickets, driven by a subset of officers that practice discrimination, in Florida. Anwar & Fang (2006) test for racial bias in police motor vehicle searches in Florida, and find no evidence of extensive racial prejudice but cannot reject a hypothesis that officers exhibit relative racial prejudice. Finally, Kleinberg et al (2018) show that risk score algorithms that include racial demographic details can produce more equitable and efficient recommendations than algorithms that omit them in an attempt to be race-blind.

Section 2 discusses the parole process in Pennsylvania and Section 3 describes and the data. Section 4 describes the empirical models and estimates for parole officer effects. Section 5 discusses the parole officer survey and the relationship between survey responses and parole officer effects. Section 6 investigates racial and gender outgroup disparities within parolee-officer relationships, and whether they appear to be driven by racial or gender bias. Section 7 concludes.

2. Background – The Parole System in Pennsylvania

Pennsylvania has the sixth highest state parole population, with nearly 50,000 individuals on parole at any given time (Pennsylvania Board of Probation and Parole, 2018). Each paroled individual falls under the jurisdiction of one of the states' 10 parole district offices, and is assigned to one of 500 community supervision agents (i.e., "officers") in the state. Each officer is assigned a mix of individuals on parole and probation, with an average active caseload of around 60-100 individuals.

A paroled individual is released into the community to serve the remainder of his sentence under community supervision. This supervision is, broadly speaking, defined by three factors. First, the individual is assigned a parole officer to oversee his supervision. Second, the individual is required to meet with his assigned officer a certain number of times each month.[2] Third, the

[2] Specifically, individuals are assigned to one of three levels of supervision based on predicted recidivism risk: minimum – one in-person meeting every three months, one collateral contact every three months (a check-in with a close member of the individual's community such as a family member, roommate, or employer); medium – one in-person meeting every month, one collateral contact every three months; or maximum – one in-person meeting every

individual is subject to conditions – such as drug testing, employment requirements, fee payments, abstaining from alcohol, curfew, program participation requirements, and restrictions on social contacts, that they must abide by at all times while on parole.[3]

Parole officers have wide discretion – even among officers in the same parole officer unit – over the tenor of the parolee-officer relationship and whether and how to sanction paroled individuals for breaking conditions. When structuring supervision, parole officers must balance the case worker aspects of their job – building rapport with paroled individuals, providing interpersonal support, connecting them with community re-entry programs and services, etc. – with the correctional officer aspects of their job – providing surveillance, highlighting the sanctions associated with future crimes, selecting sanctions in response to violations, etc. Regarding sanctions, when an officer observes a parole violation they have substantial discretion over whether to provide a written warning, assign additional parole conditions, require a short-term stay in a non-secure facility, or re-incarcerate. Finally, parole officers can choose whether and how to be flexible regarding when and where a paroled individual must meet with them, and when and whether to assign new or remove existing parole conditions.

In Pennsylvania between 2006 and 2020 paroled individuals were assigned to regional parole officer units based on two criteria – the location of the individual's proposed residency and the required special needs of the paroled individual, such as alcohol and other drugs needs, sexual offender needs, domestic violence offender needs, and mental health needs. Within each officer unit, paroled individuals were quasi-randomly assigned to parole officers based on officer caseload size at the time of assignment. Specifically, within each parole officer unit paroled individual were assigned to the parole officer in the unit with the smallest caseload at the time of assignment.

From 2006 to 2020 individuals were assigned to parole officers several months before release. At this time the officer would visit and vet the individual's proposed residency (i.e., their

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two weeks, one collateral contact every month. For all levels of supervision, every other in-person contact must be at the individual's approved residence. If the individual is in a treatment program, the officer must conduct double the required number of collateral contacts, such that every other contact is with the treatment provider. Collateral contacts may be in-person, over the phone, or over email. Additionally, while most parole supervision levels are determined by recidivism risk, sex offenders and individuals convicted of domestic violence are automatically assigned maximum supervision, and individuals released to community correction centers are assigned no less than medium supervision while in residence. After one year on parole, based on good behavior individuals are eligible for a substantially lower level of supervision (e.g., administrative supervision, special circumstance supervision, and monitored supervision).

[3] Parole conditions are initially assigned by parole hearing parole interviewers. After release, parole officers have discretion to impose additional conditions in response to parole violations or remove conditions in response to good behavior.

"home plan"). If approved, the individual was released into the community to the approved residential address under the supervision of the assigned parole officer who approved the plan. If denied, the individual could propose an alternative residence. If the residence falls within the bounds of a different parole officer unit, the individual was reassigned to an officer in that new unit. This new officer would then visit and vet the new proposed residency, and if approved would oversee the individual's supervision upon release. If at the time of release no proposed residency had been approved, the individual was released to a Community Corrections Center (CCC) (i.e., "halfway house"). Specifically, the individual was released to the CCC – with an available room – located closest to their most recent proposed residency, and assigned to one of the parole officers associated with that CCC. Officer assignment is quasi-random at CCC's as well – individuals are assigned officers based on either officer caseload size at the time of release or, for certain CCCs, the quasi-random location of their assigned room in the CCC.[4]

3. Data

Data on convictions and prison stays, home plans, parole violations, and parole officer-documented employment comes from the Pennsylvania Department of Corrections (DOC). Data on arrests is provided by the Pennsylvania State Police (PSP), and data on tax record-documented employment (from employer unemployment compensation quarterly tax reports) is provided by the Pennsylvania Department of Labor & Industry (L&I).[5]

Table 1 provides information on individuals paroled to home plans in Pennsylvania, at the time of release. Between January 1st, 2006, and January 1st, 2019, 50,839 individuals were released to parole under home plans.[6] In Pennsylvania, the majority of individuals living in the community under parole supervision are male, and the population is fairly evenly split between

[4] Note that paroled individuals are often reassigned to new officers at a later date, due to individuals changing addresses, parole officers changing units or going on leave, or specific parole officer inner-office decisions and requests. In this analysis I focus on initial officer assignment and evaluate intent-to-treat effects. As such, note that the observed officer effect will be a lower bound of the total officer effect if no reassignments took place. Secondly, after one year on parole without incident individuals are eligible for a lower level of "administrative parole" in Pennsylvania. As such I focus on parole officer effects during the first one-year period of parole.

[5] Each data set is available between 3/06 and 12/19. Specific data set availability ranges are as follows: convictions and incarcerated stays (1/04-6/23); parole hearings (1/04-12/19); home plans (1/05-8/25); parole violations (3/06-6/23), parole officer-documented employment (3/06-3/21); arrests (1/04-5/22); and tax record-documented employment (1/05-12/19).

[6] Appendix Table A1 provides summary statistics on the 51,500 individuals paroled to Community Corrections Centers between 31/2006, and 1/1/2019. Overall, a higher percentage of this population was convicted of violent crimes and has higher recidivism risk than individuals paroled to home plans.

non-Hispanic white and Black individuals. Educational attainment and marriage rates are quite low – 35% of individuals have not completed a high school degree, and only 16% are married at the time of release. Convicted crime type is evenly split between violent crimes, drug crimes, and property crimes, and the average remaining sentence time at release is 3.5 years.

Table 1, Panel B provides information on one-year post-release outcomes. On average, nearly one-third of individuals paroled to home plans return to state custody within one year of release, via a combination of arrests and technical parole violations (i.e., violations of parole conditions that trigger re-incarceration). Nearly 40% receive at least one minor parole violation – i.e., violations of parole conditions that trigger a written warning, new conditions, or placement in a non-secure community facility – and only around half are ever documented as employed during the first year after release.

Figure 1 provides more granular detail on post-release outcomes, over a three-year time horizon. As shown in Panel A, recidivism rates spike in the second quarter after release then slowly fall over time. This reduction is mostly driven by a reduction in technical parole violations that trigger re-incarceration, as individuals complete their convicted sentences and move off of parole supervision over time. Panel B shows a similar trend for minor parole violations, which also spike in the second quarter after release then slowly fall over time.

For paroled individuals, employment rates are similar, on average, when measured using state wage-record data or parole officer-documented employment data. However, these average rates mask notable differences in employment trends between the datasets, as shown in Figure 1 Panel C. Theoretically, both employment datasets are potentially biased with the biases pointing in opposite directions. For state wage records, individuals and employers have incentives to underreport employment to reduce tax obligations. For parole officer-documented employment, individuals have incentives to overreport employment to comply with the parole condition, assigned to nearly all individuals, to be actively working (or searching for work) at all times. The data depicts these biases via opposing employment trends during the first three years after release – state wage record-documented employment rates decrease each quarter (from 43% to 32%) while parole officer-documented employment rates increase each quarter (from 33% to 44%). These trends fit a hypothesis that, over time, more individuals begin to (1) receive their pay under the table (by transitioning from formal employment arranged before release to informal employment) and (2) convince their parole officers that they are employed regardless of actual employment (as,

anecdotally, parole officers become more strict about the "actively working" parole requirement over time).

Figure 1, Panel D provides state wage data on industry of employment and wages during the first three years after release. Upon release from prison, among employed individuals, 22% are employed in temporary help positions, 16% in food services, 15% in manufacturing, 10% in construction, and 8% in retail trade. Over the first three years after release the percent employed in temporary help positions decreases (to 11%) while the percent employed in manufacturing and construction slightly increase (to 19% and 13%, respectively). Finally, total wages are exceptionally low, but increasing (from $3,400 to $5,800 per quarter) among recently paroled, employed individuals.

Finally, Table 2 breaks down the sanctions that are given for each type of parole violation in Pennsylvania. For the 50,839 individuals paroled to home plans between 2006 and 2018, 43% of parole violations are for drug test failures, 26% are for broken parole conditions, 23% are for arrests, and 7% are for absconding. Arrest violations almost always result in reincarceration.[7] Absconding violations result in reincarceration two-thirds of the time, new conditions 20% of the time, and written warnings 11% of the time. Broken condition violations are usually sanctioned with written warnings and new conditions, only resulting in reincarceration occasionally (10% of the time). Finally, drug test failures are sanctioned with new conditions nearly half of the time, written warnings 24% of the time, non-secure facility stays 16% of the time, and reincarceration 14% of the time.

4. Parole Officer Effects

Research Design

There are many ways to evaluate the extent to which "as-good-as-randomly" assigned individuals – such as police officers, teachers, or supervisors – impact outcomes. My primary specification leverages the literature on randomized decision makers (e.g., Bhuller et al., 2020; Dobbie et al., 2018; Agan et al., 2021; Weisburst, 2024; LaForest-Tucker, 2026).[8] This approach

[7] Arrests that do not lead to reincarceration are for low-level summary offenses, i.e. non-traffic citations such as disorderly conduct, harassment, or shoplifting.

[8] For other examples of "randomized decision-maker" designs see, for example, Bhuller et al. (2020), Dobbie et al. (2018), Bhuller et al. (2018), Mueller-Smith (2015), Loeffler (2013) (judges in criminal court settings), Agan et al. (2021) (prosecutors in a criminal court setting), Weisburst (2024) (police officers to calls for service), and Gross & Baron (2021) (child welfare investigators to child maltreatment investigations).

creates "leave-one-out" expected effects for each officer, which can be used to test both the "relevance" of officer effects on post-release outcomes and the "exogeneity" of officer assignments.

For this specification, as the model includes exogenous covariates (i.e., parole officer units), I construct officer effect measures using Unbiased Jackknife Instrumental Variables Estimation (UJIVE) (Kolesár, 2013). UJIVE eliminates potential biases from the inclusion of covariates, ensures they are handled consistently, and remains consistent even with a large number of covariates (Chyn et al., (2024)). To create the jackknifed officer effect measure for the parole officer assigned to individual (j), for outcome Y, I first estimate outcome predicted values for every paroled individual other than (j) between 2006 and 2019, that take into account both officer fixed effects (O_{it}) and assigned parole officer unit-year fixed effects (U_{it}). These values, $PredY_{it}$, are constructed as the predicted values from an OLS regression of the equation

$$Y_{it} = \gamma_0 + \gamma_1 O_{it} + \gamma_2 U_{it} + e_{it} \quad , \qquad (2)$$

using all observations $i \neq j$, where e_{it} is a stochastic error term.[9] Next, I partial out U_{it} from these predicted values, to create an unbiased measure of officer outcome effect. Specifically, the officer effect for paroled individual (j) – V_{jt} – is constructed as the predicted residuals from an OLS regression of the equation

$$PredY_{it} = \gamma_0 + \gamma_1 U_{it} + e_{it} \quad , \qquad (3)$$

using all observations $i \neq j$.

Note that, in addition to varying how they provide supervision, officers may vary in their propensity to approve proposed residences. Even if officers are quasi-randomly assigned prior to release, if approval rates vary than officer assignments at time of release may no longer be random. Appendix Table 1 tests the relevance and exogeneity of initially assigned agent home plan approval effects. While initial assignments appear to be random (a formal joint f-test for independence between the leave-one-out officer home plan approval effects and other observable characteristics about the individual (X_{it}) produces a p-value of 0.56), officers appear to approve home plans at different rates (the f-value for the relationship between the leniency measure and

[9] The primary specification drops (1) officers assigned less than 20 total paroled individuals and (2) officer unit-year combinations where only one officer was ever assigned.

home plan approval, controlling for all other observable characteristics about the individual, is 411).[10]

If officer home plan leniency is correlated with officer outcome effects, the subset of individuals released to home plans may be unbalanced across officers. Due to this concern I estimate two primary specifications. The first assumes no such bias and considers randomization groups U_{it} to be comprised of only parole officer unit and year. The second directly incorporates officer home plan leniency into randomization groups U_{it}.[11] This specification creates exogenous "outcome effect" measures that compare individuals in the same parole officer unit and year, who were quasi-randomly assigned officers with similar home plan approval leniencies. This specification recovers causal, unbiased estimates of the relative effect of parole officers on post-release outcomes (Humphries et al., 2025; LaForest-Tucker, 2026). Note, however, that controlling for officer home plan leniency in randomization group fixed effects U_{it} (which are now substantially more granular) reduces the available sample size, as individuals in randomization groups U_{it} without officer variation are dropped.

An alternative specification – to evaluate the extent to which parole officer assignment impacts post-release outcomes – is to compare the extent to which individual officer fixed effects impact outcome estimates, relative to other observable characteristics about the paroled individual (Weisburst, 2024). Specifically, let Y_{it} be the outcome for individual i after release at time t (reincarceration, parole violation, or employment), let X_{it} be a set of observable personal characteristics about the paroled individual and level of parole supervision, let U_{it} be the paroled individual's parole officer unit and year released, and let ε_{it} be a stochastic error term. The impact of observable personal characteristics on post-release outcomes, without taking into account individual parole officer effects, is given by vectors β_1 and β_2 in Equation 1:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \varepsilon_{it} \quad . \qquad (1)$$

[10] X_{it} is comprised of the following covariates: male, black, Hispanic, married, divorced / separated, high school graduate, age at release, violent crime conviction, drug crime conviction, public order crime conviction, minimum sentence length, whether paroled at minimum sentence hearing, remaining sentence length at release, medium recidivism risk, and high recidivism risk.

[11] The primary analysis splits interviewer parole leniency into three bins - lenient, neutral, and strict - in randomization groups W_{it}. As shown in Appendix Figure A8, results are appreciable similar for alternative leniency group specifications.

Next, let $Agent_i^j$ equal one if individual i is assigned to parole officer j at the time of release, and zero otherwise. The impact of observable personal outcomes when taking into account parole officer effects is given by vectors β_1, β_2, and β_{3j} for each parole officer j in Equation 2:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \sum_j \left(\beta_{3j} Officer_i^j\right) + \varepsilon_{it} \quad . \qquad (2)$$

Additionally, note that the adjusted R^2 associated with an OLS regression of Equation 1 (AR_1^2) conveys the percent of variation in outcomes that can be ascribed to observable characteristics not including parole officer. Note, additionally, that the adjusted R^2 associated with an OLS regression of Equation 2 (AR_2^2) conveys the percent of variation in outcomes that can be ascribed to observable characteristics including parole officer. As such, $AR_2^2/AR_1^2 - 1$ represents the percent of total observed variation in outcomes attributable to assigned parole officers.

Results

Results for relevance and exogeneity tests are presented in Table 3. Panel A presents relevance and exogeneity tests for minor parole violation effect measures. For both specifications, the measures pass exogeneity tests - formal joint f-tests for independence between the measures (V_{jt}) and other observable characteristics about the individual (X_{it}) produce p-values of 0.32 and 0.99, respectively. For both specifications, the parole violation effect measures are also quite relevant - first stage f-values for the relationship between the expected parole violation effect measure and whether a parole violation occurred, controlling for all other observable characteristics about the individual (X_{it}), are 447 and 174, respectively.

Panel B presents the results from relevance and exogeneity tests for each other outcome of interest. When creating effect measures controlling for officer home plan leniency (Specification B), the measures appear exogenous from all other observation characteristics about the individual, with p-values between .03 and .99 with an average of .42. When creating effect measures without controlling for officer home plan leniency (Specification A) exogeneity results are less strong, with p-values between .00 and .86 with an average of 0.16.

The relevance of officer effect measures varies significantly across outcomes. Officers appear to have little effect on arrests (F-stats of 7 and 4), a slight effect on parole violation reincarceration (F-stats of 51 and 18), and a substantial effect on each type of minor parole violation sanction (F-stats ranging from 83-446). Interestingly, officers have a substantial effect

on officer-documented employment (F-stats of 822 and 367) but almost no effect on tax-record-documented employment (F-stats of 11 and 4).

These relevance results are shown graphically in Figure 2. Each graph presents a histogram of officer leave-one-out effects for a different outcome, along with a local linear regression (LLR) of an individual's outcome on assigned parole officer leave-one-out effects for that outcome. While all histograms appear normally distributed, the local linear regressions for arrest and tax-record-documented employment are nearly flat, conveying that these officer expected effects have little impact on these outcomes. Conversely, the local linear regressions for minor parole violations and officer-documented employment show much stronger relationships. For each, the LLRs are close to monotonically increasing with relatively high slopes. Specifically, moving from the officer with the highest expected parole violation effect to the lowest expected parole violation effect leads to a 25 percentage point decrease (from 50% to 25%) in the likelihood of incurring a minor parole violation. Similarly, moving from the officer with the lowest expected officer-documented employment effect to the highest expected officer-documented employment effect leads to a 25 percentage point increase (from 37% to 62%) in the likelihood of ever being documented as employed during the first year after release.

Results for the "officer fixed effects" regressions reach the same conclusion, as shown in Table 4. Columns 1 and 2 present the Adjusted R^2's for regressions of different outcome variables, over the first year after release to parole, excluding and including officer fixed effects, respectively. As shown in Column 1, all known information about an individual on parole, other than assigned parole officer, can only explain 6-12% of post-release outcomes. Controlling for randomly assigned parole officer increases R^2 by somewhere between zero and four percentage points, depending on the outcome, as shown in Column 4. Additionally, Column 3 presents the percent of adjusted R^2 attributable to the officer fixed effects. Overall, officers appear to have little relative effect on post-release recidivism, only explaining 3% and 9% of the explainable variation in arrest and parole violation reincarceration outcomes, respectively. However, officers appear to have a large relative effect on documented minor parole violations, that results in a written warning or new restrictions but do not result in reincarceration, explaining 24% of the explained variation in the first year after release. While officers have a large relative effect on officer-documented employment (explaining 36% of the explained variation) they have little effect on state-tax-record documented employment (explaining only 2% of the explained variation).

These results can be interpreted in several ways. First, it does not appear that individual parole officer choices or relationships have a large effect on keeping released individuals out of prison, either by decreasing future arrests or parole violations that are serious enough to warrant re-incarceration (e.g., absconding or failing a drug test). Second, while it does appear that individual officers do have an effect on minor parole violations, it is unclear what is driving these effects. For example, certain officers may build relationships with paroled individuals that cause them to change their post-release behavior, impacting their propensity to break minor conditions of parole. However, it is also possible that these officers have no effect on behavior. Instead, it is possible that the observed effects in the data are driven by differences in officer reporting decisions. For example, certain officers may be more likely to write up and sanction individuals for violations, even if the officers have no direct impact on the behavior that causes those violations. As such, these results could be driven by actual officer effects on parolee behavior, differences in officer reporting behavior, or a combination of both.

The employment results suggest that officer outcome difference are driven by differences in reporting behavior. As officers have a large effect on officer-documented employment, but little effect on tax-record-documented employment, these results bolster a hypothesis that differences in officer behavior is not impacting violation behavior or an individuals propensity to find and keep work once released to the community. Instead, these results fit a hypothesis that certain officers are more likely to follow up and vet whether parolee reported employment is factual, and more likely to write up and sanction individuals for violations.

Finally, do officers that increase documented employment also decrease minor parole violations? To investigate this I compare the leave-one-out measures for these outcomes, and find a correlation between the documented employment effect and documented minor violation effect of .10. This correlation implies that assignment to an officer that increases documented employment also appears to increases documented minor violations, at least to an extent.

5. Interpersonal Support, Structural Support, and Deterrence

I next investigate the relationship between parole officer effects and officers' self-reported approaches to parole supervision, as documented by a survey of parole officers. In Pennsylvania, the Department of Corrections surveyed all parole officers in three consecutive years – 2015, 2016, and 2017. The survey included 37 questions and statements about parole officers' job

perspectives, such as "Reentry is part of supervision", "I build rapport with the parolees on my caseload", "I want to help parolees", and "I consider myself more of a case worker than a police officer", scored on a scale of 1 (strongly disagree) to 5 (strongly agree).[12] While parole officers were not required to complete the survey, response rates were fairly high – on average 59% of parole officers responded to the survey each year.

The 37 questions broadly fall into 8 categories. The first three cover officers' perspectives on the interpersonal relationships between themselves and paroled individuals. "*Good Relationship*" questions refer to whether officers prioritize having a good relationship with their parolees, good rapport, and whether they think that their parolees feel comfortable talking to them about issues. *"Care about Parolees"* questions refer to the extent to which they care about paroled individuals and want to help them, and "*Learn about Parolees*" questions refer to the extent they try to learn about paroled individuals and think about things from their points of view. The next two categories cover officers' perspectives on their role of providing structural support. "*Help Parolees Change Behavior*" questions refer to whether officers try to help parolees learn pro-social skills, navigate the re-entry process, and provide mentoring, and "*Social Work is Part of Job*" questions refer to the extent officers prioritize the social work aspects of their job over the law enforcement aspects. Finally, "*Favorable Opinion of Parole Policies*" questions refer to whether officers think current Pennsylvania Parole policies help with reentry and whether current workloads are manageable, and "*Never Responded*" is an indicator variable for whether the officer never responded to the survey.

Let S_i represent the vector of survey responses for the parole officer assigned to paroled individual i. The relationship between parole officer perspectives S_i and individual i's outcome Y_{it} is represented by β_s in Equation 6,

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \beta_s S_i + \varepsilon_{it} \quad , \qquad (6)$$

where U_{it} is the paroled individual's assigned parole officer unit-year and X_{it} is a vector of other observable characteristics about the individual. Given that the survey was only conducted from 2015 and 2017, I do not allow the effect of survey response to vary over time. For parole officers who filled out the survey in multiple years only the first completed survey is used.[13]

[12] See Appendix B for the full set of survey questions.
[13] Results are appreciably similar if I instead use the most recent completed survey.

Summary statistics for aggregate survey categories are presented in Table 5.[14] On average, officers that responded to the survey care about building interpersonal relationships with paroled individuals – responding with an average of 3.9 out of 5 across interpersonal relationship-related questions. Officers also prioritize providing skill-building and structural support to paroled individuals, with an average of 3.8 across questions about providing structural support. Finally, officers that responded to the survey had a somewhat negative opinion of parole policies, with an average of 2.5 across questions that ask about whether workload sizes are manageable and parole policies are supportive.

The impact of parole officer job perspectives, as captured by survey responses, on outcomes is also presented in Table 5. Generally, I find little effect of survey responses on arrests, supporting the conclusion that assigned parole officers appear to have little relative effect on arrests in Pennsylvania. As before, I observe an effect of parole officers on minor parole violations. Specifically, officers that care about building good relationships reduce minor violations, those that prioritize helping paroled individuals change behavior reduce both major and minor violations, and those that believe that social work is a major part of their job increase minor violations. Specifically, a one standard deviation increase in survey response to questions about whether having a good relationship with a parolee is important, or whether they help a parolee after the parole makes a mistake are associated with 3.6 and 2.3 percentage point decreases in minor parole violations in the first year after release.

However, the results suggest that individuals who are assigned to officers who focus on providing structural support receive more minor parole violations than individuals who are assigned to officers who do not focus on providing structural support. For example, a one standard deviation increase in survey response to questions about whether social work is part of a parole officer's job is associated with a 2.8 percentage point increase in minor parole violations in the first year after release. Finally, parolees assigned to officers who believe current PBPP policies are reasonable have a 3.9 percentage point higher parole officer-documented employment rate, and parolees assigned to officers who never responded to the optional survey have a 3.7 percentage point higher officer-documented employment rate.

These results show that individuals assigned to parole officers that focus on building interpersonal connections have better post-release outcomes related to minor parole violations.

[14] Results and summary statistics for each individual survey question are presented in Appendix Table A3.

Even if the result is driven not by paroled individual behavioral change, but by officer reporting behavior, reducing minor parole violations is still beneficial for the paroled individual – these violations often lead to additional parole restrictions that, in themselves, do not improve individual outcomes (LaForest-Tucker, 2026) but do increase the burden of supervision. As such, they motive a conclusion that officers that focus on providing interpersonal support produce better post-release outcomes than officers that do not.

It is not clear whether the effect of interpersonal support are driven by differences in job perspectives themselves, or something about certain officers that cause them to have certain job perspectives. For example, these effects could be caused by the job perspectives themselves. If this is the case, then these results lead to a policy recommendation centered on parole officer training - more officers should be trained to prioritize interpersonal connections. Alternatively, however, these effects could be driven by unobserved characteristics of certain parole officers, which they bring to the job with them when they are hired and which happen to be correlated with their job perspectives. If this is the case, then these results lead to an policy recommendation centered on parole officer hiring – that more officers that exhibit an interest in building interpersonal connections should be hired.

6. Outgroup Disparities and Bias

Finally, I investigate the relationship between parole officer demographic characteristics, paroled individual demographic characteristics, and outcome effects. First, the relationship between parole officer observable characteristics, P_{it}, and assigned paroled individual outcomes is represented by β_p in Equation 7,

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \beta_p P_{it} + \varepsilon_{it} \quad , \qquad (7)$$

where P_{it} is comprised of parole officer race, gender, and years of service variables.

The relationships between parole officer demographic characteristics and assigned paroled individual outcomes are presented in Table 6. Being assigned a male or black officer has no effect on recidivism or tax-record-documented employment, but each leads to a decrease in minor parole violations and parole-officer documented employment, relative to being assigned a female or white officer. Additionally, individuals assigned to officers with more years of service are less likely to have major or minor parole violations, and more likely to be documented by their officer as employed, during the first year post-release. Specifically, individuals assigned officers with one

additional year of service are 1.0% less likely to have a parole violation that results in reimprisonment, and 0.6% less likely to have a minor parole violation that leads to a written warning or additional parole conditions. These results fit a hypothesis that female officers, white officers, and newer officers may be assigning more conditions in order to build up authority and agency with their assigned paroled individuals.

I next explore the impact of differences in relative parolee-officer race and gender. Let Q_{it} represent a vector of the relative race and gender quadrant indicators of the parolee-officer relationship for paroled individual i ("white-white" / "white-nonwhite" / "nonwhite-nonwhite" / "nonwhite-white"; "male-male" / "male-female" / "female-female" / "female-male"). The effect of relative parolee-officer race and gender is represented as β_q in Equation 8,

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \beta_q Q_{it} + \varepsilon_{it} \quad . \tag{8}$$

Results on the average effect of relative parolee-officer race, and relative parolee-officer gender, are presented in Table 7.[15] While differences in officer approaches to interpersonal support, structural support, and deterrence have little effect on recidivism, being assigned a parole officer of the opposite race (white vs. Black or Hispanic) has a notable impact on recidivism. Specifically, white individuals assigned to non-white officers have a 7% higher recidivism rate, on average, than white individuals assigned to white officers, after taking into account parole officer unit and all other observable characteristics about each parolee. Additionally, non-white individuals assigned to white officers have an 11% higher recidivism rate than non-white individuals assigned to non-white officers, leading to a net difference in recidivism of being assigned to an officer of the opposite race of 18%. While not statistically significant, the results also suggest that being assigned an officer of the opposite race leads to increased minor parole violations and decreased officer-documented and tax-record-documented employment.

I also observe notable effects of relative parolee-officer gender. Male individuals appear to have similar levels of recidivism regardless of whether they are assigned to male of female parole officers, but female individuals have an 8% lower recidivism rate when assigned to female parole officers. Conversely, female individuals appear to have similar levels of minor parole violations regardless of whether they are assigned to male of female officers, but male individuals have an 6% higher minor violation rate when assigned to female officers. Additionally, while

[15] Detailed estimate coefficients are presented in Appendix Table A5.

female officers increase officer-documented employment for both male and female individuals, they increase it substantially more for female individuals (9% and 19%, respectively). Finally, while female officers have no impact on tax-record documented employment for men, they increase tax-record documented employment by 10% for women.

These results suggest aggregate-level evidence of outgroup effects in parolee-officer relationships. Note, however, that these effects could be driven by a variety of factors. They could be driven by parole officer racial and gender bias. For example, some non-white officers could be treating white individuals more harshly than they should be, or black individuals more leniently than they should be, or some white officers could be treating white individuals more leniently, or black individuals more harshly, than they should be. Any of these four hypothesis are consistent with the observed outgroup disparities. Alternatively, these disparities could be driven by how individuals themselves *respond* to being assigned parole officers of a different race. That is, individuals might respond differently to officers of different races, even if the officers themselves treat all assigned individuals the same way. Finally, these effects could be caused by a joint mix of these two effects, if perceived differences in race impact the strength and quality of the supervisory relationship parole officers and assigned individuals are able to, jointly, construct. It is unclear from these aggregate estimates whether the effects are driven by bias among certain parole officers, bias in parolee behavioral responses themselves, or combined structural bias that impacts the relationships that parolees and parole officers are able to construct.

To investigate these potential mechanisms, I explore the extent to which aggregate-level racial and gender outgroup disparities in parolee-officer pairings are observable at the individual-level. To do so, I construct measures of individual level differential effects across individual race and gender subgroups. First, I estimate parole officer effects on each subsample of individual – white parolees, non-white parolees, male parolees, and female parolees – following Equations 3-5. I then define the individual-level "white/non-white relative effect" of each parole officer as the difference in that parole officer's effect on white individuals relative to their effect on non-white individuals. Similarly, I define the individual-level "male/female relative effect" of each parole officer as the difference in that parole officer's effect on male individuals relative to their effect on female individuals.

Distributions of these effects are displayed in Figure 3. Figure 3, Panel A presents the distribution in the white parolee/non-white parolee relative effect of each parole officer (top

histogram), each white parole officer (middle histogram), and each non-white parole officer (bottom histogram), on arrests, minor parole violations, and tax-documented employment. These figures show no evidence of outlier officers with very large white parolee / non-white parole relative effects - the distribution of these effects for both white officers and non-white officers appear to be close to normally distributed. However, the means of these distributions are slightly different. Specifically, white officers have a mean slightly below non-white officers, signifying that white parolees assigned to them have a slightly lower propensity to incur minor parole violations, in the aggregate, than non-white parolees assigned to them, relative to outcomes for individuals assigned to non-white officers. The same is true, in the reverse direction, for tax-documented employment – white parole officers have slightly higher average documented employment numbers for white parolees than non-white parolees, relative to non-white parole officers. The same results are apparent for officers and paroled individuals of different genders, as shown in Figure 3, Panel B. Once again, these figures show no evidence of outliers, the distributions appear to be close to normally distributed, and the only difference between the distributions is a slight shifting of the mean.

Overall, these results are consistent with the hypothesis that observed racial and gender bias in parolee – parole officer relationships is not caused by "a few bad apples." The bias observed in the aggregate data appears to be fairly evenly distributed across officers, consistent with a hypothesis that these effects are driven by larger, systemic or structural biases that impact the way parolees and parole officers of different races and genders are able to trust and build relationships together.

7. Discussion

This work shows that individual parole officers do not appear to have a notable relative effect on parolee recidivism – whether through future arrests or parole violations that lead the reincarceration. However, individual officers do appear to have an effect on minor parole violations – that lead to written warnings or additional parole conditions – as well as documented employment. These effects are also important, as minor parole violations often lead to additional conditions which a paroled individual must abide by, and post-release employment rates are particularly low for previously incarcerated individuals. Based on parole officer survey responses, officers that prioritize interpersonal supports are by-and-large the same officers that reduce

violations, providing suggestive evidence that parolee-officer relationships that focus on interpersonal supports can be beneficial for paroled individuals. However, officers that prioritize structural supports do not appear to have a notable effect on parolee outcomes, providing suggestive evidence that these types of existing support services are not effective in facilitating reentry.

This work also finds evidence of racial and gender bias in parolee-officer relationships. Specifically, being assigned a parole officer of a different race or gender leads to slightly higher rates of recidivism, higher rates of minor parole violation, and lower employment. However, there is no evidence this bias is driven by individual officers. Instead, the evidence supports a hypothesis that these relative effects are driven by structural or societal biases that negatively impact the ability of a parolee and parole officer to build a strong and supportive relationship. As such, the results motive a conclusion that these biases must be tackled at a more aggregate societal scale. However, parole officers should be aware of these potential biases so that they can actively work to reduce them when building parolee-officer relationships.

Overall, these results support a conclusion that the parolee-officer relationship has a nuanced effect on the outcomes of paroled individuals. Changing the behavior of certain officers – and employing more officer that prioritize building interpersonal connections with paroled individuals – could reduce minor parole violations and improve documented employment. However, the conclusion from this work reinforce the results of other work on the effects of community corrections and reentry programs – that these programs and supervision policies appear to have little impact on returns to custody. As such, more research is needed to test alternative programs and policies that could appreciably improve the reentry outcomes of paroled individuals.

Tables & Figures

Table 1 – Paroled Population

	Mean
A. Demographic Characteristics	
Male	88%
Black	38%
Hispanic	8%
Married	16%
Divorced / Separated	11%
No High School Degree	35%
Age at Release	35.0
Violent Crime Conviction	25%
Drug Crime Conviction	34%
Property Crime Conviction	20%
Minimum Sentence Length (Years)	2.5
Paroled at Minimum Sentence Hearing	67%
Remaining Sentence at Release (Years)	3.5
Medium Recidivism Risk Level	42%
High Recidivism Risk Level	29%
B. Outcomes – One Year After Release	
Recidivism *(Arrested or Reincarcerated)*	29%
Arrest	20%
Parole Violation Reincarceration	10%
Minor Parole Violation	39%
Ever Employed (Parole Data)	55%
Ever Employed (L&I Data)	61%

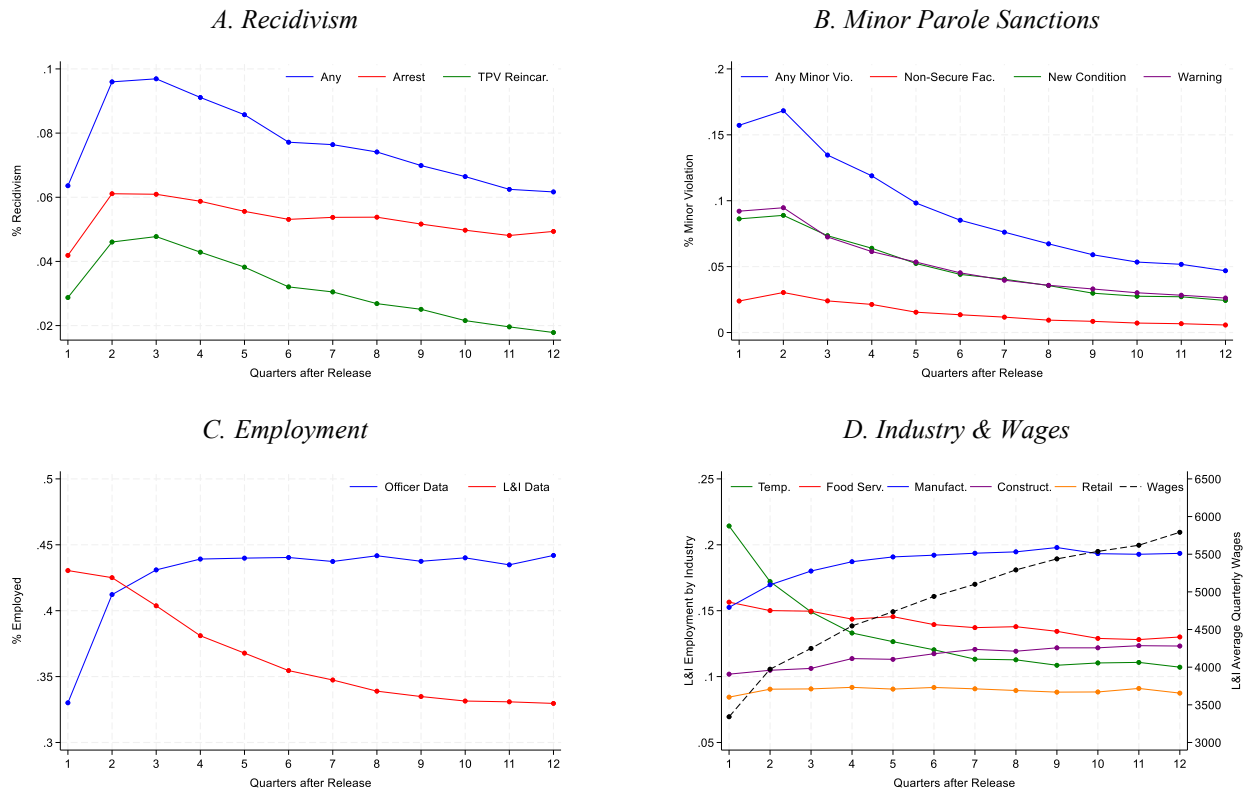
Notes: This table presents summary statistics for the 50,839 home plan releases to parole in Pennsylvania between 3/1/2006 and 1/1/2019.

Table 2 – Parole Violations & Sanctions

	Sanction Type				
	Written Warning	New Conditions	Non-Secure Facility Stay	Reincarc.	**Total**
Violation Type					
Broke Condition	51%	31%	4%	10%	38,281
Failed Drug Test	24%	46%	16%	14%	64,862
Absconded	11%	20%	4%	64%	9,511
Arrested	3%	14%	2%	81%	34,588
Assaultive Behavior	4%	24%	4%	67%	1,415
Weapon Possession	14%	21%	6%	59%	663
Total	37,058	48,631	12,841	48,306	

Notes: Percentages aggregate from left to right. This table presents summary statistics for all parole violations that occur before 6/1/2023 for the 50,839 home plan releases to parole in Pennsylvania between 3/1/2006 and 1/1/2019. N = 149,344.

Figure 1 – Outcomes by Quarter



Notes: Figure depicts three years of average post-release outcomes, by quarter, for the 39,835 home plan releases to parole in Pennsylvania between 3/1/2006 and 1/1/2017 . Graph A presents arrest and non-arrest technical parole violation reincarceration outcomes. Graph B presents minor parole violation outcomes, broken out by minor violation type – non-secure facility stays, new conditions, and written warnings. Graph C compares quarterly employment rates from parole officer-documented employment data and PA Department of Labor and Industry-documented employment data. Graph D presents industry-specific employment rates and average wages - among the subset of employed individuals - from PA Department of Labor and Industry-documented employment data. N = 39,835 for all outcomes except parole officer-documented employment. Note that parole officer-documented employment sample sizes decrease from Q1-Q12 (N = 36,455 in Q4, 27,730 in Q8, and 20,002 in Q12) as some individuals reach their max sentence date and are no longer observed on parole.

Table 3A – Relevance & Exogeneity Summary for Officer Outcome Effect Measures

	Primary Outcome Propensity Measures				**Propensity Measures Conditional on Home Plan Leniency**			
	Relevance Test		***Exogeneity Test***		***Relevance Test***		***Exogeneity Test***	
	F-Stat.	*P-Val.*	*F-Stat.*	*P-Val.*	*F-Stat.*	*P-Val.*	*F-Stat.*	*P-Val.*
Recidivism (*arrested or reincar.*)	8.0	*(.00)*	2.6	*(.00)*	1.3	*(.25)*	1.4	*(.14)*
Arrest	7.4	*(.00)*	2.0	*(.01)*	3.5	*(.06)*	1.0	*(.43)*
PV Reincarceration	51.0	*(.00)*	1.5	*(.08)*	17.1	*(.00)*	1.4	*(.10)*
Minor Parole Violation	447.0	*(.00)*	1.1	*(.32)*	173.8	*(.00)*	.4	*(.99)*
Sanction Type								
Written Warning	446.1	*(.00)*	0.7	*(.86)*	269.3	*(.00)*	.5	*(.98)*
New Condition	312.3	*(.00)*	1.6	*(.04)*	112.8	*(.00)*	.9	*(.61)*
Non-Secure Facility Stay	84.1	*(.00)*	1.5	*(.10)*	82.8	*(.00)*	1.2	*(.29)*
Ever Employed (Parole Data)	822.2	*(.00)*	2.4	*(.00)*	367.2	*(.00)*	1.2	*(.24)*
Ever Employed (L&I Data)	11.0	*(.00)*	2.2	*(.00)*	4.3	*(.04)*	1.7	*(.03)*

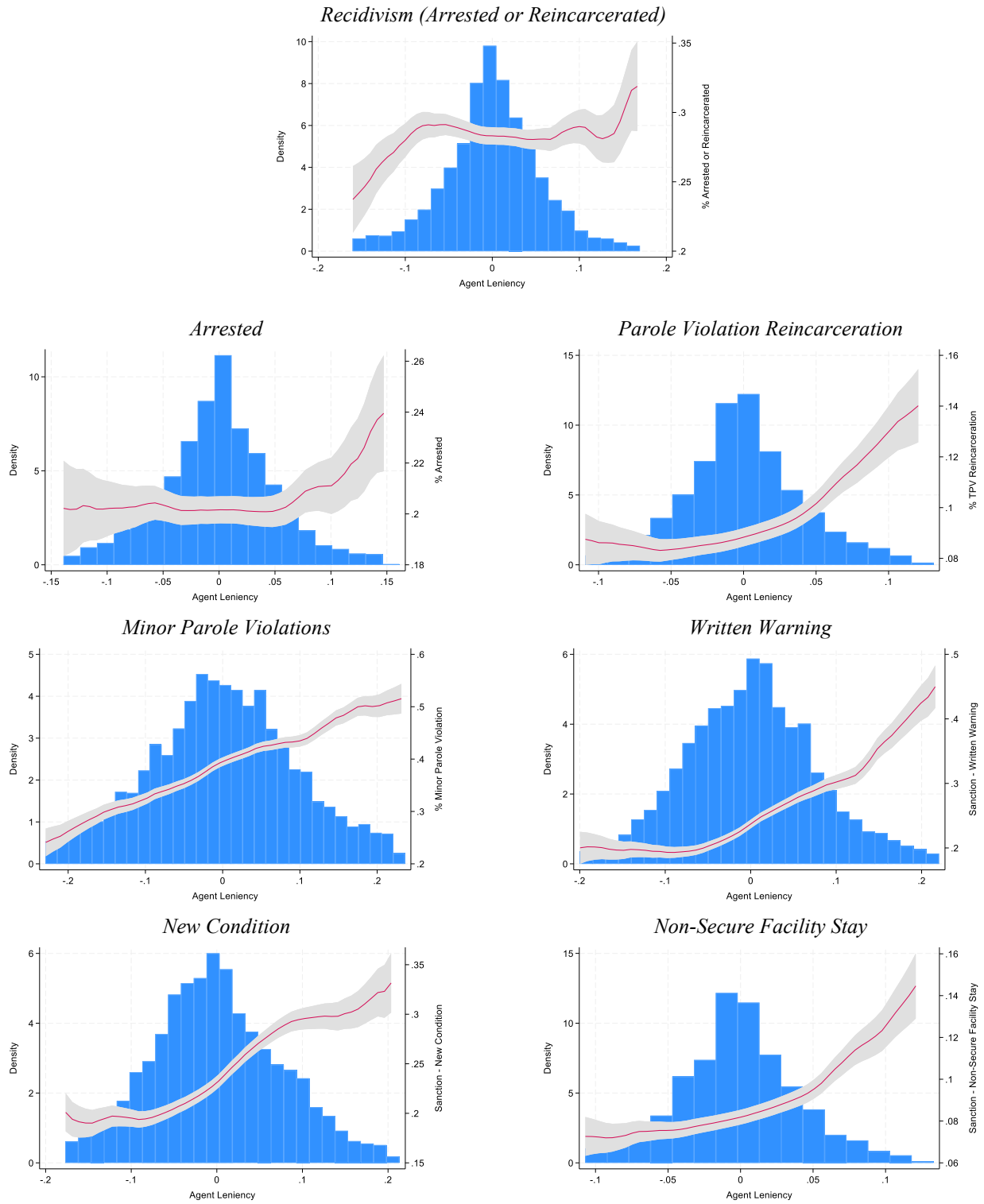
Notes: Columns 1 & 3 present f-test values for whether β(propensity) = 0, from regressions of whether each outcome occurred during the first-year post-release on each officer outcome propensity measure, controlling for observable characteristics about the individual. Columns 2 & 4 present joint f-test p-values from regressions of officer outcome propensity measures on observable characteristics about the individual. When constructing officer minor violation propensity, Columns 3 & 4 include three agent home plan leniency types (lenient, neutral, and strict) in randomization groups U_{it}, and drop randomization groups for which there is no longer sufficient officer variation. N = 29,062 for Columns 1 and 2 and N = 22,017 for Columns 3 and 4.

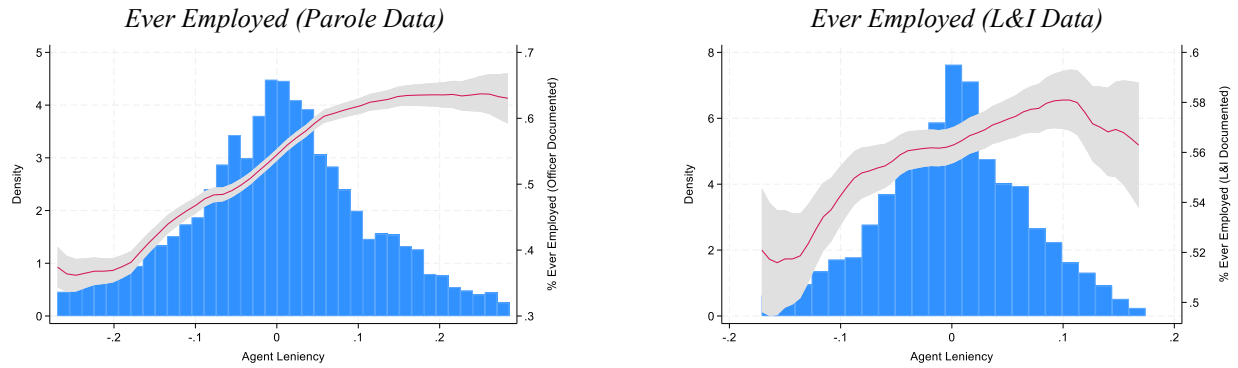
Table 3B – Relevance & Exogeneity Details for Officer Minor Parole Violation Effect Measure

	Primary Outcome Propensity Measure				**Propensity Measure Conditional on Home Plan Leniency**			
	Relevance Test		***Exogeneity Test***		***Relevance Test***		***Exogeneity Test***	
	Est.	***SE***	***Est.***	***SE***	***Est.***	***SE***	***Est.***	***SE***
Dependent Variable	***Minor Violation Occurred***		***Minor Violation Propensity***		***Minor Violation Occurred***		***Minor Violation Propensity***	
Minor Violation Propensity	.5166 ***	*.0244*	-	-	.3759 ***	*.0284*	.0011	*.0026*
Male	.0151 *	*.0091*	-.0012	*.0023*	.0137	*.0106*	-.0007	*.0023*
Black	.0285 ***	*.0080*	-.0022	*.0020*	.0304 ***	*.0093*	.0031	*.0034*
Hispanic	.0095	*.0120*	-.0003	*.0029*	.0127	*.0138*	-.0002	*.0023*
Married	-.0327 ***	*.0080*	-.0033 *	*.0020*	-.0337 ***	*.0093*	.0019	*.0029*
Divorced / Separated	.0000	*.0100*	-.0036	*.0025*	-.0101	*.0117*	.0000	*.0017*
High School Graduate	-.0164 ***	*.0059*	.0010	*.0014*	-.0169 **	*.0068*	.0000	*.0001*
Age at Release	-.0041 ***	*.0003*	.0000	*.0001*	-.0044 ***	*.0003*	-.0006	*.0026*
Violent Crime Conviction	-.0812 ***	*.0092*	.0020	*.0023*	-.0854 ***	*.0107*	.0007	*.0024*
Drug Crime Conviction	-.0592 ***	*.0084*	.0051 **	*.0021*	-.0556 ***	*.0097*	.0000	*.0006*
Minimum Sentence Length (Years)	-.0114 ***	*.0020*	.0003	*.0005*	-.0125 ***	*.0023*	.0004	*.0018*
Paroled at Minimum Hearing	-.0112 *	*.0063*	.0004	*.0016*	-.0137 ***	*.0073*	.0001	*.0004*
Remaining Sentence (Years)	-.0029 **	*.0014*	-.0002	*.0003*	-.0018	*.0016*	.0026	*.0020*
Medium Recidivism Risk Level	.1153 ***	*.0072*	.0007	*.0018*	.1122 ***	*.0083*	.0024	*.0022*
High Recidivism Risk Level	.1956 ***	*.0079*	.0010	*.0019*	.1926 ***	*.0091*	.0011	*.0026*
Test F-Statistic	447.0		1.1		*174.8*		*0.4*	
P-Value	*(.00)*		*(.32)*		*(.00)*		*(.99)*	

Notes: Columns 1 & 3 present f-test values for whether β(propensity) = 0, from a regression of whether a minor violation occurred during the first year post-release on officer minor violation propensity, controlling for observable characteristics about the individual. Columns 2 & 4 present joint f-test p-values from a regression of officer minor violation propensity on observable characteristics about the individual. When constructing officer minor violation propensity, Columns 3 & 4 include three agent home plan leniency types (lenient, neutral, and strict) in randomization groups U_{it}, and drop randomization groups for which there is no longer sufficient officer variation. N = 29,062 for Columns 1 and 2 and N = 22,017 for Columns 3 and 4.

Figure 2 – Variation in Officer Effects





Notes: Figures present histograms of parole officer effect measures, along with local linear regressions of the outcome of interest on the UJIVE effect measure. N = 29,062 releases to home plans in parole officer units with quasi-random officer assignment.

Table 4 – Alternative Specification – Officer Fixed Effects

	Adjusted R^2 w/o Officer Fixed Effects	Adjusted R^2 with Officer Fixed Effects	% Obs. Variation Due to Officer	% Total Variation Due to Officer
Recidivism	0.086	0.089	3%	0%
Arrest	0.063	0.065	3%	0%
PV Reincarceration	0.077	0.084	9%	1%
Minor Parole Violation	0.110	0.136	24%	3%
Sanction Type				
Written Warning	0.124	0.149	20%	3%
New Condition	0.080	0.101	26%	2%
Non-Secure Fac. Stay	0.063	0.072	14%	1%
Ever Emp. (Parole Data)	0.120	0.163	36%	4%
Ever Emp. (L&I Data)	0.097	0.099	2%	0%

Notes: Table presents adjusted R^2 measures from regressions of whether each outcome occurred during the first-year post-release on observable characteristics about the individual and randomization groups U_{it}. N = 29,062 releases to home plans in parole officer units with quasi-randomly assigned officers.

Table 5 – Relationship between Officer Effects and Survey Results

		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed - L&I Data	
Survey Categories	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Interpersonal Support											
Good Relationship	3.87	.001	*.009*	.001	*.006*	-.036***	*.010*	-.010	*.011*	-.021**	*.010*
Care about Parolees	3.99	-.002	*.011*	-.004	*.008*	.001	*.013*	.001	*.014*	-.008	*.014*
Learn about Parolees	3.74	-.008	*.009*	.014**	*.007*	.012	*.011*	-.014	*.011*	.006	*.011*
Skillbuilding / Structural Support											
Help Parolees Change Behavior	4.03	.013	*.009*	-.014**	*.006*	-.023**	*.010*	-.005	*.011*	.007	*.011*
Social Work is Part of Job	3.70	-.015*	*.008*	.006	*.006*	.028***	*.009*	.011	*.010*	.019**	*.010*
Other Categories											
Favorable Opinion of Par. Policies	2.45	-.001	*.004*	-.003	*.003*	-.004	*.005*	.039***	*.005*	.012**	*.005*
Responded	0.68	-.002	*.007*	-.012**	*.005*	-.016*	*.008*	.037***	*.009*	-.008	*.008*

Notes: Table presents estimates from regressions of whether each outcome occurred during the first-year post-release on each aggregate officer survey measure, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 26,083 releases to officers who were emailed the survey, and N = 17,653 releases to officers who responded to the survey, among parole officer units with quasi-random officer assignment.

Table 6 – Aggregate Effects of Officer Demographic Characteristics

		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed - L&I Data	
	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Male Officer	0.77	.006	*.007*	.000	*.005*	-.019**	*.008*	-.061***	*.009*	-.007	*.008*
Black Officer	0.15	-.015	*.009*	-.003	*.006*	-.014	*.011*	-.036***	*.011*	-.001	*.011*
Hispanic Officer	0.04	-.039***	*.015*	-.002	*.011*	.020	*.017*	.022	*.018*	-.009	*.018*
Officer Years of Service	9.86	.001**	*.000*	-.001***	*.000*	-.002***	*.001*	.001**	*.001*	-.001	*.001*

Notes: Table presents estimates from regressions of whether each outcome occurred during the first-year post-release on each officer demographic characteristics, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 29,062 releases to parole officer units with quasi-random officer assignment.

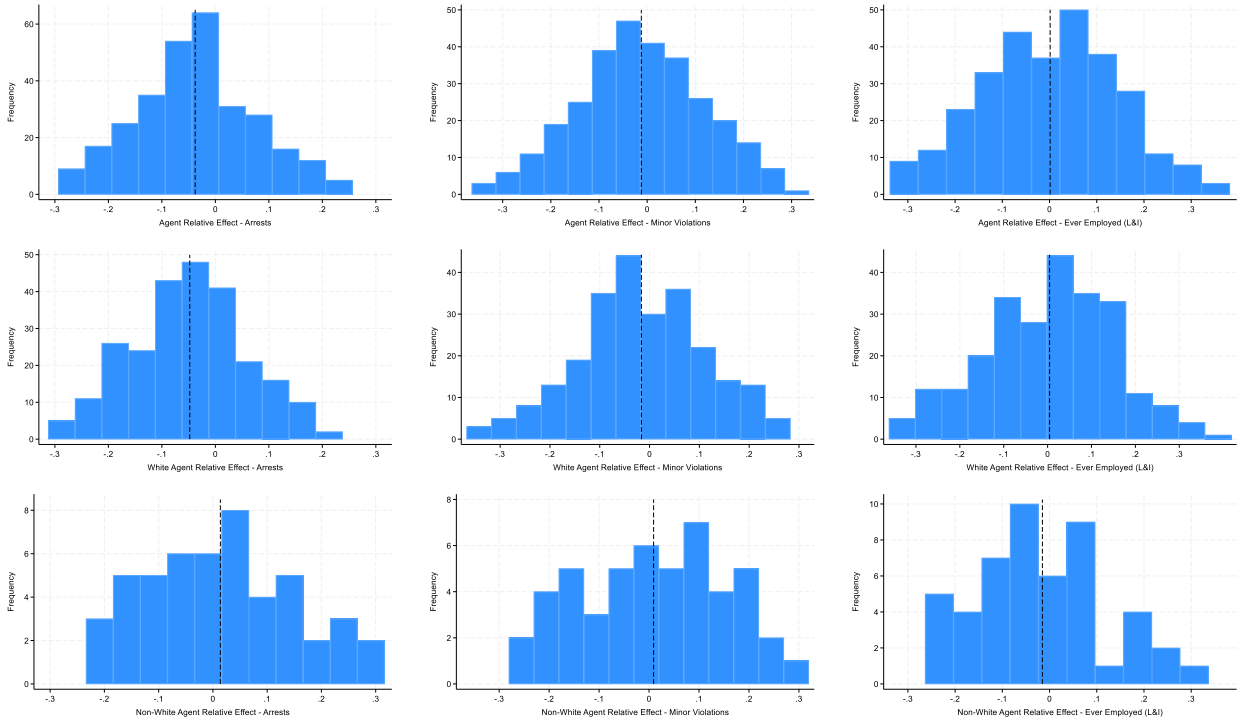
Table 7 – Outgroup Disparities

	Racial Outgroup Disparities				Sex Outgroup Disparities		
	White Officer	Non-White Officer	*Overall Effect - Opposite Race*		Male Officer	Female Officer	*Overall Effect - Opposite Sex*
Recidivism							
White Parolee	0	7%	18%***	Male Parolee	0	1%	9%
Non-White Parolee	10%	-1%***		Female Parolee	-24%	-32%	
Arrest							
White Parolee	0	11%	24%***	Male Parolee	0	2%	10%
Non-White Parolee	22%	9%***		Female Parolee	-21%	-30%	
PV Reincarceration							
White Parolee	0	-4%	5%	Male Parolee	0	3%	5%
Non-White Parolee	-28%	-38%		Female Parolee	-27%	-29%	
Minor Parole Violation							
White Parolee	0	-2%	3%	Male Parolee	0	6%***	4%
Non-White Parolee	4%	-1%*		Female Parolee	-3%	-1%	
Ever Employed (Parole Data)							
White Parolee	0	-6%*	-3%	Male Parolee	0	9%***	-10%**
Non-White Parolee	-17%	-21%*		Female Parolee	-16%	4%***	
Ever Employed (L&I Data)							
White Parolee	0	-4%	-4%	Male Parolee	0	0%	-10%**
Non-White Parolee	-5%	-6%		Female Parolee	1%	11%***	

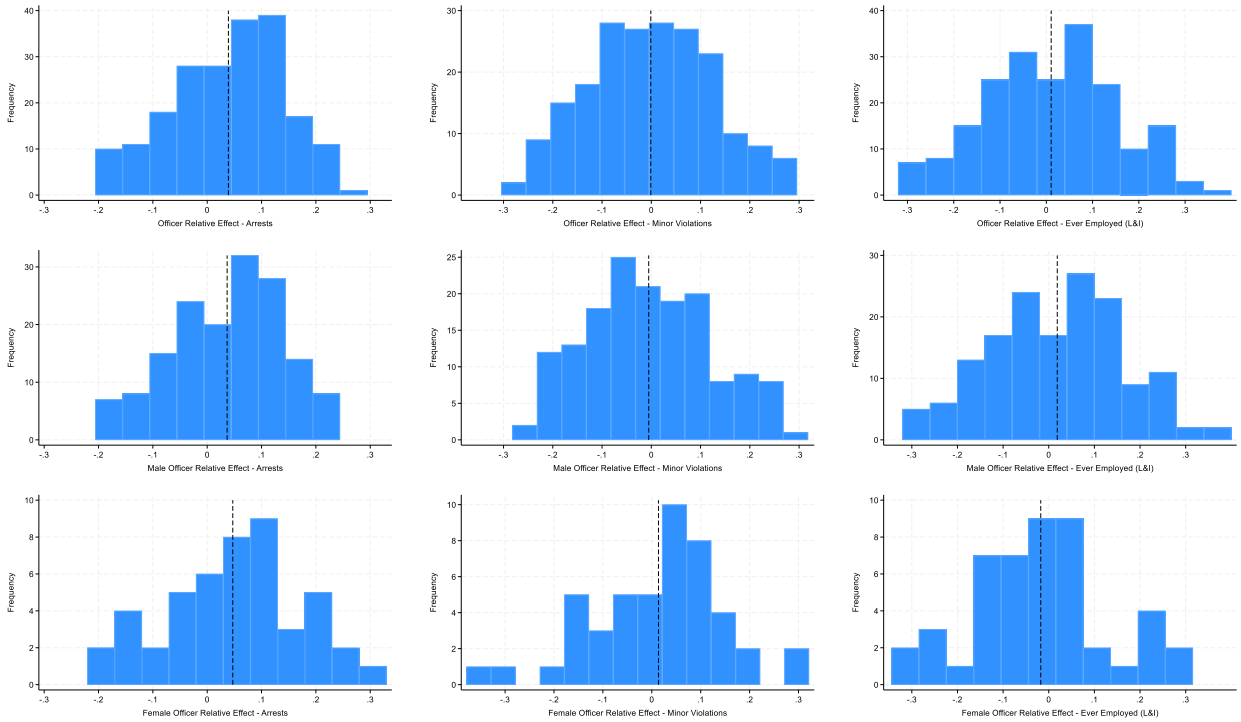
Notes: Table presents estimates from regressions of whether each outcome occurred during the first-year post-release on indicators for relative parolee-officer race and gender, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 29,062 releases to parole officer units with quasi-random officer assignment.

Figure 3 – Relative Officer Outgroup Effects

Panel A – Difference in Officer Effect for White vs. Non-White Paroled Individuals



Panel B – Difference in Officer Effect for Male vs. Female Paroled Individuals



Notes: Panel A presents the distribution in the white parolee/non-white parolee relative effect of each parole officer (top histogram), each white parole officer (middle histogram), and each non-white parole officer (bottom histogram), on arrests, minor parole violations, and tax-documented employment. Panel B presents the distribution in the male parolee/female parolee relative effect of each parole officer (top histogram), each male parole officer (middle histogram), and each female parole officer (bottom histogram), on arrests, minor parole violations, and tax-documented employment.

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Appendix A – Auxiliary Tables and Figures

Table A1 – Community Corrections Center Releases

	CCC Releases (Mean)	Home Plan Releases (Mean)
A. Demographic Characteristics		
Male	92%	88%
Black	48%	38%
Hispanic	12%	8%
Married	13%	16%
Divorced / Separated	10%	11%
No High School Degree	43%	35%
Age at Release	35.2	35.0
Violent Crime Conviction	30%	25%
Drug Crime Conviction	32%	34%
Property Crime Conviction	22%	20%
Minimum Sentence Length (Years)	3.0	2.5
Paroled at Minimum Sentence Hearing	55%	67%
Remaining Sentence at Release (Years)	3.7	3.5
Medium Recidivism Risk Level	36%	42%
High Recidivism Risk Level	43%	29%
B. Outcomes – One Year After Release		
Recidivism *(Arrested or Reincarcerated)*	39%	29%
Arrest	21%	20%
Parole Violation Reincarceration	23%	10%
Minor Parole Violation	41%	39%
Ever Employed (Parole Data)	47%	55%
Ever Employed (L&I Data)	62%	61%

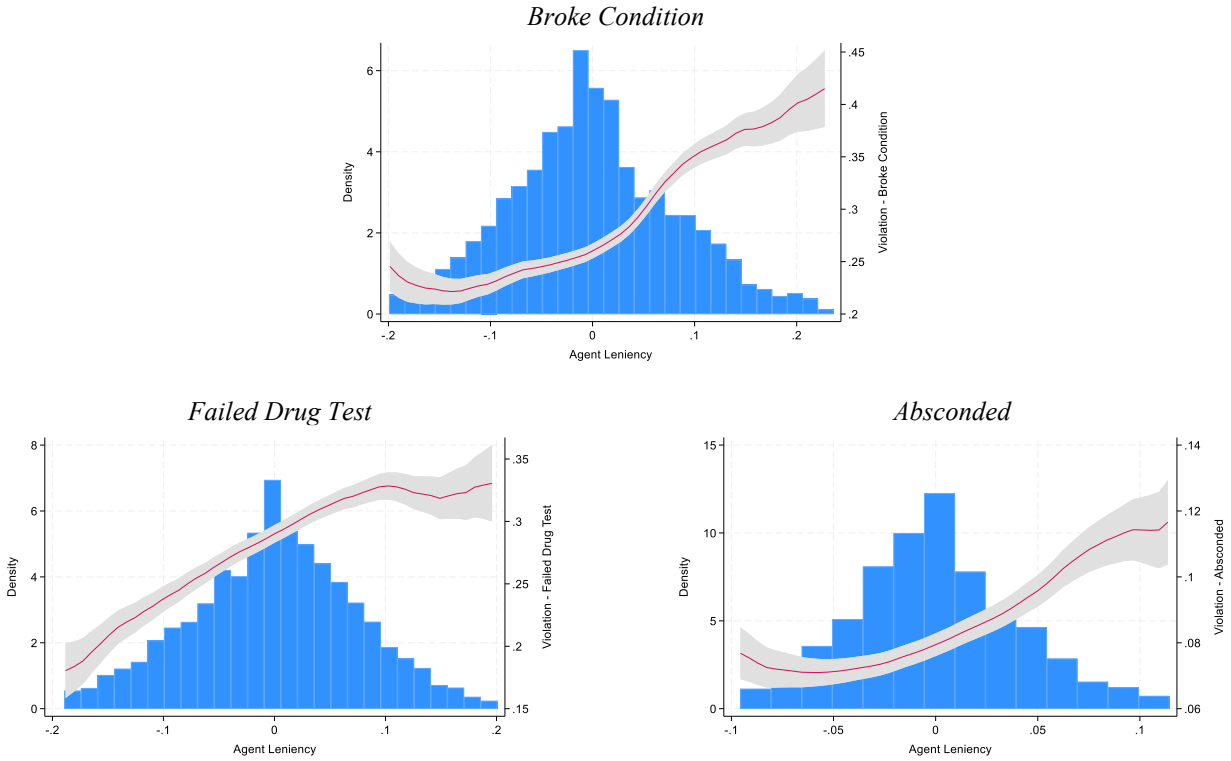
Notes: This table presents summary statistics for the 50,839 home plan releases to parole (Column 1), and 51,500 Community Corrections Center releases to parole (Column 2), in Pennsylvania between 3/1/2006 and 1/1/2019.

Table A2 – Home Plan Instrument Validity

	Relevance Test		***Exogeneity Test***	
	Est.	***SE***	***Est.***	***SE***
Dependent Variable	***HP Approved***		***HP Leniency***	
Home Plan Leniency	.5011 ***	*.0247*	-	-
Male	-.0292 ***	*.0072*	.0005	*.0014*
Black	-.0807 ***	*.0058*	.0004	*.0012*
Hispanic	-.0314 ***	*.0090*	.0002	*.0018*
Married	-.0047	*.0059*	-.0006	*.0012*
Divorced / Separated	-.0149 **	*.0074*	-.0020	*.0015*
High School Graduate	.0403 ***	*.0043*	.0012	*.0009*
Violent Crime Conviction	-.0188 ***	*.0067*	.0000	*.0013*
Drug Crime Conviction	.0277 ***	*.0063*	.0000	*.0013*
Minimum Sentence Length (Years)	.0028 ***	*.0008*	-.0003 *	*.0002*
Test F-Statistic	411		0.87	
P-Value	.00		.56	

Notes: Column 1 presents f-test values for whether β(leniency) = 0, from a regression of home plan outcome on agent home plan leniency, controlling for observable characteristics about the individual. Column 2 presents joint f-test p-values from a regression of agent home plan leniency on observable characteristics about the individual. N = 43,210 initial home plan investigations in parole officer units with quasi-randomly assigned officers.

Figure A1 – Variation in Officer Effects – Violation Types



Notes: Figures present histograms of parole officer effect measures, along with local linear regressions of the outcome of interest on the UJIVE effect measure. N = 29,062 releases to home plans in parole officer units with quasi-random officer assignment.

Table A3 - Relationship between Officer Effects and Individual Survey Question Responses

		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed - L&I Data	
Survey Questions	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Promote Prosocial	4.05	.003	*.008*	.002	*.006*	.005	*.009*	.015*	*.009*	.008	*.009*
Rapport Important	4.07	-.003	*.006*	.004	*.005*	.025***	*.007*	.024***	*.008*	-.010	*.008*
Know Parolee Situation	4.18	-.003	*.007*	.006	*.005*	-.007	*.008*	-.016*	*.009*	.000	*.008*
Good Relationship	4.03	.003	*.008*	-.002	*.006*	-.018**	*.009*	-.011	*.009*	-.015	*.009*
Current Policies Supportive	2.90	.000	*.004*	-.006*	*.003*	-.010*	*.005*	.016***	*.006*	.008	*.005*
Help Success Society	4.11	.002	*.008*	-.005	*.006*	-.001	*.009*	-.011	*.010*	-.013	*.010*
Mentor Parolees	3.50	-.002	*.005*	-.003	*.004*	-.014**	*.006*	-.009	*.006*	.001	*.006*
Parolees Can Learn	3.69	.001	*.006*	-.006	*.005*	.008	*.007*	-.007	*.008*	-.008	*.007*
Workload Achievable	2.00	-.003	*.005*	.004	*.004*	.010	*.006*	.029***	*.006*	.008	*.006*
Case Worker	3.31	-.004	*.004*	.000	*.003*	.006	*.004*	.005	*.004*	.005	*.004*
Supervision Balance	3.75	-.007	*.005*	.004	*.003*	.023***	*.005*	.014**	*.006*	.007	*.006*
Reentry Important	3.95	-.003	*.008*	.000	*.006*	.008	*.009*	.023**	*.010*	-.008	*.010*
Care Parolees	3.97	-.006	*.007*	.008	*.005*	.003	*.008*	-.010	*.009*	-.006	*.008*
Optimistic with Parolees	3.66	.001	*.006*	-.013***	*.005*	.002	*.007*	.007	*.008*	-.005	*.008*
Parolees Comfortable	3.80	-.003	*.007*	-.005	*.005*	-.028***	*.008*	-.007	*.009*	-.006	*.008*
Treat Fairly	4.69	.004	*.010*	-.001	*.008*	.007	*.012*	.035***	*.013*	-.017	*.012*
Praise for Success	4.23	.002	*.008*	-.013**	*.006*	-.013	*.009*	-.001	*.010*	.012	*.010*
Want to Help	4.24	.009	*.008*	.005	*.006*	-.006	*.010*	-.014	*.010*	.012	*.010*
Consider Parolee Views	3.76	-.003	*.008*	-.003	*.006*	.015	*.009*	.032***	*.010*	-.013	*.010*
Think about Reason Behavior	3.74	-.003	*.006*	.000	*.005*	.002	*.008*	-.004	*.008*	.004	*.008*
Work with Parolees	3.86	.002	*.008*	.001	*.006*	-.003	*.009*	.004	*.009*	.003	*.009*
Nonjudgemental	4.14	-.008	*.007*	-.003	*.005*	-.016**	*.008*	-.002	*.008*	-.007	*.008*
Programming Match Risk	3.54	.009	*.006*	-.003	*.004*	.003	*.007*	-.008	*.007*	.004	*.007*
Identify Triggers	3.69	.006	*.006*	.009*	*.005*	.016**	*.008*	.006	*.008*	.001	*.008*
Contact Support System	3.78	-.019***	*.007*	.012**	*.005*	.009	*.008*	-.005	*.008*	.018**	*.008*
Demonstrate Prosocial Skills	4.22	-.001	*.006*	-.005	*.005*	.016**	*.007*	.008	*.008*	.003	*.007*
Interventions Match Risk	3.29	-.004	*.006*	.011***	*.004*	-.001	*.007*	-.011	*.007*	.007	*.007*
I Build Rapport	4.29	.009	*.009*	.009	*.006*	-.018*	*.010*	-.003	*.011*	.015	*.011*
Provide Warning	3.99	.006	*.006*	-.005	*.005*	-.013*	*.008*	-.017**	*.008*	-.002	*.008*
Explain Consequences	4.24	.006	*.008*	-.007	*.006*	.000	*.009*	-.007	*.009*	.002	*.009*
Parolees Initially Motivated	3.52	.004	*.005*	-.002	*.004*	-.002	*.006*	.005	*.006*	.002	*.006*
Parolees Seem Hopeful	3.47	.001	*.006*	.006	*.005*	.014*	*.008*	-.016**	*.008*	.004	*.008*
Skill Building Discussions	3.07	.011*	*.006*	.006	*.004*	-.004	*.007*	-.006	*.007*	.004	*.007*
Set Parolee Goals	3.71	-.008	*.007*	-.006	*.006*	.005	*.009*	.017*	*.009*	-.016*	*.009*
Identify Criminogenic Needs	3.90	-.001	*.006*	-.005	*.005*	.002	*.007*	-.021***	*.008*	.003	*.008*
Parolees Can Change	3.51	-.003	*.005*	.004	*.004*	-.005	*.006*	.004	*.007*	.008	*.007*
Parolees Talk Issues	3.15	-.003	*.005*	-.001	*.004*	-.005	*.006*	-.014**	*.007*	-.004	*.007*

Notes: Table presents estimates from regressions of whether each outcome occurred during the first-year post-release on each officer survey response, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 26,083 releases to officers who were emailed the survey, and N = 17,653 releases to officers who responded to the survey, among parole officer units with quasi-random officer assignment.

Appendix B - Parole Officer Survey Questions

"In my opinion…"

Possible responses - Strongly Agree, Agree, No Opinion, Disagree, Strongly Disagree

1. Promoting pro-social behavior is an important part of my job as a parole officer/parole supervisor.
2. Building rapport with parolees is important to their success under supervision.
3. Knowing each parolee's current situation makes my job easier.
4. Having a good relationship with a parolee makes my job easier.
5. Current Parole Board policies are supportive of offender rehabilitation.
6. One of my roles as a parole officer/parole supervisor is to help parolees become successful members of society.
7. One of my roles as a parole officer/parole supervisor is to mentor parolees.
8. Parolees are unable to change their negative behaviors.
9. Parolees are capable of learning prosocial ways of thinking.
10. The workload demands set forth by Central Office are achievable.
11. Select "Strongly agree" here.
12. I consider myself more of a case worker than a police officer.
13. Supervision should be balanced between social work and law enforcement.
14. Reentry is part of supervision.

"Consider your interactions with parolees and choose the most representative response:…"

Possible responses - Almost Always, Frequently, Sometimes, Occasionally, Hardly Ever

15. I care about parolees as people.
16. I am optimistic with parolees.
17. Parolees seem to feel comfortable enough to be open and honest with me.
18. I praise parolees for the good things they do.
19. I want to help parolees.
20. I consider parolees' views.
21. Parolees seem to keep important issues to themselves and won't tell me about them.
22. I think about the underlying reasons for parolee behavior.
23. When a parolee makes a mistake, I work with him to find out what brought him to that point.
24. I use a nonjudgmental approach in my interactions with parolees.
25. I match programming to a parolee's risk level.
26. I help parolees identify triggers and develop plans to effectively address those triggers.
27. I have frequent contact with parolees' support systems.
28. I demonstrate prosocial skills before asking a parolee to do so.
29. I am able to use risk level to apply appropriate interventions to each parolee.
30. I build rapport with the parolees on my caseload.
31. When parolees are going in a bad direction, I talk with them before taking serious action.
32. If a parolee breaks the rules, I calmly explain the consequences of his actions.

33. At the first contact, right after release from prison, parolees seem motivated to change.
34. Prior to maxing out of supervision, parolees seem hopeful about their future.
35. I include skill building discussions in each meeting with parolees.
36. I work with parolees to set goals for their personal improvement.
37. I can identify a parolee's criminogenic needs.